Service Portfolio

Semantic Governance

Service Portfolio

Vocabulary Management

- Text analysis, term extraction
- Semantic association
- Assign attribute/properties (nodes & edges)
- Ability to categorize edges
- Ability to mine new terms from imported RDFs, OWL structures
- Assign weights to edges (esp. for rules)
- Reasoning & Inference Rules (RIF) language (leveraging terms from SPIN or SHACL)
- Instantiate math symbolic (calcs, rules, etc.)
- Constructing consistent URIs, IRIs
- Need a universal identifier that we can rely upon for identifying all data (and things) separate from their labels & properties

Taxonomy Management

- Building taxonomies
    - 🌕 Creating SKOS-based vocabularies
    - 🌕 Auto tagging & semantic extension
    - 🌕 Organizing entities into hierarchies
    - 🌕 Building associated taxonomies
- Managing taxonomies
    - 🌕 Inheritance support (direct & indirect*)
    - 🌕 *polymorphic
- Taxonomy Lifecycle Management
    - 🌕 change management
    - 🌕 version management
    - 🌕 workflow and control
    - 🌕 Impact analysis (for changes)

- 🌕 publish / promotion
- 🌕 Warning messages if moving or pruning nodes

Ontology Management

- Defining patterns in ontologies
  - Ability to evolve taxonomies into a KG
- NLP supported search
- Deprecation w/o deletion
- Declarative Query Language (SPARQL, GraphQL, Cypher, ..)
- Transaction Management
  - 🌕 ACID support is key
- Augmented search
- Navigate structured and unstructured data
- Harmonize & query content from multiple sources
- Evolutionary development & engineering
- Mapping engineering
  - 🌕 Ontology matching / alignment
  - 🌕 Schema matching / alignment
  - 🌕 Semantic matching / alignment

Relationship Management

- Meta property (attributes, properties for relationships)
- Event relations (n-ary support)
- Information extraction from entities and relations
  - 🌕 Inferred meaning
  - 🌕 disambiguation

Semantic Governance

Service Portfolio

Lifecycle Management

- Data Governance
    - Managing Nulls & duplicates in data
    - Data Transformation (privacy)
    - Masking, Obfuscation, Tokenization, Encryption
    - Matching, Fusion, Merging
- Social Engineering & Communication
    - Bridge domain expertise
    - Bridge technical expertise
- Support and enforce adherence to methodology
    - Knowledge capture
        - Discovery – vocabulary – taxonomy - ontology
    - Implementation
        - Validation – mapping – query – measurement
- Lifecycle Management
    - Capture inputs, processing, outputs across lifecycle
    - Ideas, Use Cases, Design artifacts, Decisions, Assumptions
    - Support for impact analysis and simulation
    - Incremental and iterative development

Performance Management

- Performance scales linearly and horizontally
- Support for centralized and distributed into graph DBs
- Support for SHACL constraints & rules construction
- Support for Location-based intelligence
- Support for on-prem & off-prem (multiple clouds)
- Includes or accommodates extensions for:
    - AI extensions

    - Data integration
    - Analytics, aggregation
    - Modeling & analysis
    - Recommendation engine
    - Semantic data integration
    - Social network analysis
- Continuity
    - Backup, restore
    - Transaction (atomic) recovery

Integration Management (import / export)

- API support bi-directional formats:
    - RDF, RDFS, OWL, XSD, SKOS, JSON
- Security support
    - LDAP, RBAC
- Integration (bi-directional) between RDB & RDF
    - RDF2RDB Mapping
    - R2RML for RDFs
- SQL Queries
    - SPARQL Queries
    - Virtual mapping (SQL2RDF)
    - D2R Optimization (for querying RDB)

Visualization Management

- Tree form
- Hierarchical
- Cloud
- Optimization

Semantic Governance
Service Portfolio

    - Number of nodes
    - Weighted nodes
    - Pathing
- Exploration
- Filtering
- NLP query

Demand Management

Operating Model
(Intake Processes)

Define WHAT needs to change:

List of changes and/or payload of changes delivered in a file.

- **Add Items:** The full definition of any new concepts, terms, or relationships being added to the taxonomy.
- **Update Items:** The unique identifier of the item being changed, along with the specific fields that have been modified (e.g., a new definition, a change in hierarchy, or new synonyms).
- **Delete Items:** The unique identifier of the item being removed. This should also include a flag to indicate if deletion cascades to child concepts.

Define the WHY covering Semantic and Contextual Metadata

This information is crucial for traceability, validation, and a deeper understanding of the change, which are cornerstones of knowledge engineering. The KE team may accept the changes on face or suggest an alternate set of changes based on design considerations and constraints in downstream systems.

- **Change Origin:** The identifier of the source system (Marketplace in this scenario). This is vital for multi-system integrations and for understanding the source of truth for a given concept.
- **Version of Origin:** A precise version identifier or timestamp of the taxonomy state in System A at the moment the change was captured. This allows Metaphactory to detect if a change is based on an outdated version of the taxonomy, preventing conflicts and overwrites.
- **Business Rationale:** A brief, human-readable description of the purpose of the change. This provides invaluable context for future knowledge analysts, auditors, and stakeholders.

Define WHO & WHEN to introduce the requested changes

This information is essential for auditability and establishing accountability, which is a key principle of good data governance.

- **User/System ID:** The unique ID of the user or automated process that initiated the change in System A. This ensures a clear chain of custody.
- **Requested Date:** This is the desired date to complete the changes for publishing
- **Change Timestamp:** The precise time and date of the change event. This is non-negotiable for any auditable system.

Operating Model
(Intake Processes)

Define Validation and Integrity Controls

Before applying any changes, Metaphactory must have a mechanism to validate them against a set of predefined rules to maintain the integrity of the taxonomy.

- **Schema Conformance:** The change set must be validated to ensure it adheres to the taxonomy's underlying schema or ontology. For example, a change that attempts to assign a concept to a non-existent parent, or create a circular relationship, should be automatically rejected.
- **Integrity Checks:** The system should perform checks to ensure that the change does not introduce logical inconsistencies or break existing rules. This is particularly important for deletions and hierarchy changes.

Conflict Resolution Strategy

In a real-world scenario, concurrent changes can occur in Metaphactory from multiple requests (and/or sources). A key input is a clear strategy for how to handle these conflicts.

- **Conflict Policy:** A flag or directive that specifies the desired behavior when a conflict is detected. This could be one of the following:
  - 🌕 **reject:** The entire change set is rejected, and an alert is sent.
  - 🌕 **overwrite:** The incoming change from System A takes precedence and overwrites the local change in System B.
  - 🌕 **flag_for_manual_review:** The change is flagged and put into a queue for a human knowledge engineer to resolve the conflict. This is often the safest default for complex business taxonomies.

These inputs provide the minimal content to support Semantic Governance. This ensures the integrity, auditability, and trustworthiness of the organization's most valuable knowledge assets.

Components

Operating Model
(Component Breakdown by ENV)

Overview

This outline covers the 3 environments where we will create and manage all of the tools, knowledge artifacts, and services. DEV includes all the components that we need to build and maintain to support the operating model. The STG environment is where we ingest Bottom-Up schemas, extend our Top-Down Knowledge, and generate relations (Mapping) between Top-Down and Bottom-Up. PRD is where we publish knowledge artifacts for downstream consumption and/or provide Semantic Search and Conversational Search (subscription)

DEV

- Vendor Tools
    - 🌕 Patches
    - 🌕 Updates
    - 🌕 Releases
    - 🌕 Customizations (VZ)
    - 🌕 Customizations (Vendor)
- Agents
    - 🌕 Vendor Agents
    - 🌕 VZ Agents*
- Services
    - 🌕 Inbound
        - Service (Bottom-Up Ontology)
        - Service (Bottom-Up Mapping Ontology)
        - Agent (Extend Knowledge)
    - 🌕 Processing
        - Agent (Domain-Specific Mapping)
        - Agent (Extending Knowledge)
        - Agent (Search & Discover) [Master / Slave]
        - Agent (Orchestration)
        - Agent (Federation)
    - 🌕 Outbound
        - Prompt (Conversational Search)
        - Agent (Prompt Sense & Decompose)
        - Agent (Contextual Relevancy)
        - Agent (Persona Relevancy)
- Platform

Operating Model
(Component Breakdown by ENV)

- Infrastructure
- OpenShift
- Compute
- Storage
- Security (SSO)

STG

- Lifecycle Management
  - Knowledge Artifacts
    - Top-Down
      - Enterprise (one set)
        - Core Vocabularies, Taxonomies, Ontologies
        - Unifying Ontology (Domain Model)
      - Domain-Specific (one for EACH Domain & Sub-Domain)
        - DS Vocabularies, Taxonomies, Ontologies
    - Bottom-Up (one for each Schema)
      - Schema Ontology
      - Mapping Ontology
  - Services
    - Inbound
      - Ingestion Pipeline (Catalog)
      - Ingestion Pipeline (Alation)
      - Ingestion Pipeline (LookML)
      - Ingestion Pipeline (Measures)
      - IngestionP Pipeline (EnGRAM)
    - Processing
      - Search & Discover
        - Calls to Domain-Specific Agents & Vector DBs
        - Calls to LLM for VZ Gemini
      - Mapping at Scale
        - Calls to Domain-Specific Agents & Vector DBs
        - Calls to LLM for VZ Gemini
      - Extending Knowledge at Scale
        - Calls to Domain-Specific Agents & Vector DBs
        - Calls to LLM for VZ Gemini
      - AI Training & Measurement
        - Provide Positive & Negative Reinforcement to Vector DBs
        - Generate Operational Monitoring & Measurement
        - Manage Defects (Lifecycle Management) and Issues (Prod)

Operating Model
(Component Breakdown by ENV)

- Security Services
  - PRS
- Monitoring & Measurement
  - Data extraction from Logs (Metaphactory & GraphDB)
  - Data Analytics & Metrics
  - Reports & Dashboards

PRD

- Knowledge Artifacts
  - Top-Down
    - Enterprise (one set)
      - Core Vocabularies, Taxonomies, Ontologies
      - Unifying Ontology (Domain Model)
    - Domain-Specific (one for EACH Domain & Sub-Domain)
      - DS Vocabularies, Taxonomies, Ontologies
  - Bottom-Up (one for each Schema)
    - Schema Ontology
    - Mapping Ontology
- Services
  - Outbound
    - Publishing
      - Business Taxonomy to Marketplace
    - Subscribing
      - Semantic Search
    - Agent (Search & Discover)
      - Agent (Federation) [Master / Slave]
      - Agent (Orchestration)
    - Prompt (Conversational Search)
      - Agent (Prompt Sense & Decompose)
      - Agent (Contextual Relevancy)**
      - Agent (Persona Relevancy)**

*** Note: Each Agent will have it’s own Vector DB (Scenarios & Training) will call the LLM**
**** Note: These provide dynamic filtering for constraining the search (precision)**

Measurement

Operating Model
(Measures)

Top-Down Knowledge Management Metrics

These metrics focus on the governance and lifecycle of your core semantic assets.

Create / Update Core Vocabularies, Taxonomies & Ontologies:

- **Vocabulary/Taxonomy/Ontology Coverage:** The percentage of key business domains, data sources, and applications covered by a managed semantic asset.
- **Adoption Rate:** The number of unique systems, teams, or users referencing the managed semantic asset per quarter.
- **Consistency Score:** The rate of new data, terms, or definitions that align with existing, approved semantic assets.
- **Update Cadence:** The average time from a business request for a new term to its publication.
- **Stale Asset Rate:** The percentage of semantic assets that have not been updated or accessed within the past year.

Create / Update Domain-Specific Vocabularies, Taxonomies & Ontologies:

- **Domain Alignment Score:** Measures how well the domain-specific assets map to the CORE assets.
- **Usage Frequency:** How often domain-specific assets are being used for tagging, enrichment, or search within their respective business units.

Semantic Services Metrics

These metrics monitor the efficiency and quality of the automated and manual processes.

Inbound Services

- **Ingestion Queue Latency:** The average time an incoming request (e.g., for a new digital asset) waits in the queue before processing begins.
- **Ingestion Throughput:** The number of digital assets ingested and processed per hour.
- **Rejection Rate:** The percentage of inbound assets that are rejected due to poor quality, lack of metadata, or other predefined rules.

Processing Services

- **Mapping at Scale (AI-Driven)**
    - 🌕 **Accuracy (Precision & Recall):**
        - **Precision:** The percentage of AI-suggested mappings that are correct.
        - **Recall:** The percentage of all possible correct mappings that the AI identified.
    - 🌕 **Confidence Score:** The average confidence level provided by the AI for its

mapping suggestions.
- **AI Suggestion Adoption Rate:** The percentage of AI-suggested mappings that are accepted without manual correction.
- **Manual Correction Rate:** The percentage of AI-suggested mappings that require manual review or correction.
- **Mapping Queue Latency:** The average time for approved AI suggestions to be applied.
- **Throughput:** The number of mappings created per hour.

- **Extending Knowledge (AI-Driven)**
  - **Knowledge Extension Accuracy:** The percentage of new facts or relations inferred by the AI that are verified as correct.
  - **AI Bias Score:** A quantitative or qualitative measure of bias in the AI's learning and suggestions. This can be complex and may require a review of source data and the output for fairness.
  - **AI Training Dashboard Monitoring:** Metrics like training time, model loss, and validation accuracy.
- **RAG-Based Learning**
  - **VectorDB Population Rate:** The number of new vectors populated into the VectorDB per hour.
  - **Retrieval Relevance:** A score for the accuracy and relevance of the retrieved context for RAG queries.

Outbound Services

- **Semantic Search as a Service**
  - **Query Response Time:** The average time to return results for a semantic search query.
  - **Search Relevance:** A measure of how well search results match the user's intent. Can be measured by click-through rates, user feedback (e.g., thumbs up/down), or the number of results reviewed.
  - **Zero-Result Rate:** The percentage of queries that return no results.
  - **Successful Query Rate:** The percentage of queries that lead to a user finding the desired information.
- **Publishing Core Vocabularies, Taxonomies & Ontologies into a Catalog:**
  - **Catalog Sync Latency:** The time from approval of a semantic asset to its publication and availability in the catalog.

Operating Model
(Measures)

🌕 **Catalog Discovery Rate:** The number of new users or systems that discover and access the published assets.

🌕 **Notification/Communication Reach:** The percentage of stakeholders or subscribers who receive and open publishing notifications.

Platform and System Monitoring

These are foundational metrics for the entire platform's health.

- **System Uptime/Availability:** The percentage of time the platform is operational.
- **Security Vulnerability Count:** The number of security vulnerabilities identified by automated scans.
- **Incident Response Time:** The average time from a system issue being identified to its resolution.
- **Resource Utilization:** Monitoring of CPU, memory, and storage usage to ensure scalability and cost efficiency.

Dashboard

Operating Model
(Value-Driven Benchmarks)

Trusted BI & AI

This section focuses on the reliability and ethical operation of the AI and data.

- **AI Accuracy and Confidence:** A line graph or gauge showing **Precision**, **Recall**, and **AI Confidence Score** for mapping and knowledge extension.
- **AI Bias Monitoring:** A simple heat map or table showing the **AI Bias Score** across different data subsets or domains.
- **Data Integrity:** A dashboard element tracking the **Ingestion Rejection Rate** and **Manual Correction Rate** to highlight data quality issues early.
- **Platform Health:** A summary of **System Uptime**, **Security Vulnerability Count**, and **Incident Response Time** to ensure the platform itself is trustworthy.

Business Literacy

This area measures how well the semantic assets are being understood and used by the business.

- **Vocabulary Adoption & Usage:** A bar chart showing the **Adoption Rate** and **Usage Frequency** of core and domain-specific semantic assets across different business units.
- **Knowledge Discovery:** A chart tracking the **Catalog Discovery Rate** to show how many new users are finding and using the published knowledge.
- **Stale Assets:** A simple metric or list showing the **Stale Asset Rate**, indicating which assets may need to be retired or updated.

Meaning at Scale

This pillar is about the efficiency and reach of your semantic services.

- **Processing Throughput:** Gauges or line charts showing the **Ingestion Throughput** and **Mapping Throughput** to demonstrate the system's ability to handle scale.
- **AI Suggestion Impact:** A donut chart showing the **AI Suggestion Adoption Rate** vs. the **Manual Correction Rate** to quantify the value of the AI.
- **Queue Management:** A metric showing the **Ingestion Queue Latency** and **Mapping Queue Latency** to ensure smooth and timely processing of requests.

Improved Decision Making

This section links the semantic services directly to business outcomes.

Operating Model
(Value-Driven Benchmarks)

- **Search Performance:** A set of graphs tracking **Query Response Time**, **Search Relevance**, and **Zero-Result Rate** to show if users can find the information they need to make decisions.
- **Knowledge Extension Value:** A qualitative or quantitative measure of how often AI-extended knowledge is used in analytical reports or decision-making processes.

Knowledge Preservation

This pillar focuses on the long-term health and maintenance of your semantic assets.

- **Lifecycle Management:** A timeline or status tracker showing the **Update Cadence** and **Catalog Sync Latency** for new and updated assets.
- **Consistency:** A metric tracking the **Consistency Score** for new data to ensure new knowledge aligns with the existing framework.

Align Business with Technology

This final section shows how the semantic governance model is a bridge between business needs and technical execution.

- **Coverage and Alignment:** A dashboard element showing the **Vocabulary Coverage** and **Domain Alignment Score** to demonstrate that the semantic assets are addressing key business areas.
- **Engagement:** A metric tracking the **Notification/Communication Reach** to ensure that stakeholders are kept informed about changes and new releases, fostering collaboration.

OPS Workbench (TS)

Operating Model
(Operational Workbench using Tom Sawyer)

Overview for setting up Tom Sawyer Perspectives to build an SG Operations Workbench that monitors, visualizes, and governs lifecycle changes to knowledge artifacts such as vocabularies, taxonomies, and ontologies in a graph database. Perspectives is uniquely designed for exactly this type of federated, dynamic knowledge graph environment, with strong integration and visualization capabilities.

Goals: Practical Outcomes

- A real-time control panel for ontology evolution and data integrity oversight.
- A semantic impact workspace showing downstream effects of vocabulary refinement.
- A governance-driven visualization suite allowing SG teams to monitor, validate, and authorize knowledge artifact changes as they occur.

Key Features that Tom Sawyer Perspectives Provides:

Unified Knowledge Graph Modeling

Tom Sawyer Perspectives includes a schema editor that lets you design or import knowledge graph models directly from your existing graph databases or RDF sources. You can bind ontology-based data (RDF, OWL, RDFS) to the schema, enabling visualization and monitoring of knowledge artifacts from multiple repositories without duplicating the data.

Real-Time Data Integration and Change Binding

Perspectives supports automatic schema extraction and real-time integration from federated data sources such as Neo4j, Amazon Neptune, Apache TinkerPop, and SPARQL endpoints. These connectors allow the workbench to display live data views of ontologies and vocabulary relationships, helping you detect and visualize changes as they propagate across taxonomies or linked models.

- Dynamic Binding: Updates in your graph database automatically refresh within Perspectives visualizations.
- Change Detection: Schema metadata can include timestamps or version identifiers so that new ontology versions or modified relationships are highlighted.

Operating Model
(Operational Workbench using Tom Sawyer)

Visualization and Impact Analysis

The platform provides rule-based, interactive graph visualizations and dependency mapping tools for analyzing ontology relationships. You can implement:

- Node-based impact graphs to see how a taxonomy update affects dependent vocabularies or instances.
- Centrality, clustering, and pathfinding analytics to show semantic influence or dependency strength.
- Overlay metadata (like version number, date, or steward) via styles or color-coding for governance oversight.

Lifecycle and Governance Monitoring

You can augment graphs with attributes such as:

- Version history
- Approval status
- Lifecycle state (draft, validated, deprecated)
- Authorized steward or domain owner

Using Perspectives dashboards and schema-linked attributes, administrators can monitor governance workflows, visualize change approval pipelines, and assess compliance against ontological or taxonomic policy guidelines. These dashboards transform abstract RDF governance rules into a live operational interface.

Operating Model
(Operational Workbench using Tom Sawyer)

OPS Workbench Architecture

HL framework for deploying TS Perspectives as an operations layer on our semantic stack:

- **Knowledge Repositories:** Neo4j / GraphDB / Amazon Neptune
- **Integration Layer:** Perspectives schema bindings and federated connectors
- **Visualization Layer:** Workbench UI built with Perspectives Designer
- **Governance Framework:** Lifecycle metadata (status, steward, policy) stored directly within RDF or external SG system
- **Event Source (Optional):** Kafka or graph triggers for asynchronous updates

OPS Workbench Configuration

Function	Implementation in Perspectives
Change Monitoring	Link version metadata to nodes/edges and use visual cues to show modifications or pending approvals
Impact Visualization	Pathfinding analysis to display dependency propagation when a concept or class changes
Artifact Versioning	Visual panels to compare ontology versions and highlight added/removed entities
Governance Workflow View	Dashboard combining schema state, change history, and steward assignments
Integration with Graph DBs	Real-time API integration to Neo4j, Neptune, or any SPARQL endpoint
Role-based Access Control	Configurable access layers to manage who can modify or approve ontology changes